

Alfredo Mungari

Machine Learning Engineer | Data Scientist | Software Engineer

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[LinkedIn](#) • [GitHub](#) • [ResearchGate](#)

PROFESSIONAL PROFILE

Machine Learning Engineer and Data Scientist with expertise in Explainable AI and Generative AI, and research experience at CNR and CI&SS Lab. MSc in Applied Computer Science (Machine Learning & Big Data) from the University of Naples "Parthenope", graduated with **110/110 Summa Cum Laude**, and BSc with **110/110 Summa Cum Laude with Special Mention**. Comfortable across the full ML pipeline, from data engineering and synthetic data generation to model interpretability, fairness analysis, and bias detection.

PROFESSIONAL EXPERIENCE

University of Naples "Parthenope" (CI&SS Lab)

Naples, Italy

Research Intern / Master's Thesis Student

December 2024 – April 2026

- Designed and implemented a 4-stage end-to-end Python pipeline for *Feature-Driven Bias Detection* on tabular classifiers, combining counterfactual-based feature importance (BoCSoR, adapted to fully categorical data) with hierarchical Association Rule Mining (FP-Growth) for global, human-readable explanations.
- Developed a two-level mining procedure (macroscopic feature-name level, microscopic value level) and a semantic classification separating *actionable* from *biased* rules involving protected demographic attributes.
- Validated the pipeline on five U.S. Census ACS 2024 benchmarks (~1.75M records) using CatBoost and MLP, characterizing geographic bias asymmetries (race-occupation entanglement, race-geography patterns) across regional subsets.
- Augmented the benchmark data with a TabDiff-based *diffusion model* for mixed numerical/categorical synthetic generation, validating distributional fidelity via Pearson/Spearman correlation analysis and fairness metrics (Equalized Odds, Error Rate Balance, Overall Accuracy Equality).

University of Naples "Parthenope" (Nola Branch)

Nola, Italy

Programming Tutor

September 2023 – July 2025

- Provided part-time academic support for the BSc in Computer Engineering and Science for Cybersecurity, assisting 30+ students per semester.
- Delivered technical guidance and code reviews on programming fundamentals, algorithms, and data structures in C and C++.

National Research Council (CNR)

Naples, Italy

Research Intern / Bachelor's Thesis Student

April 2022 – September 2022

- Developed *AUDO4RIAS*, an open-source Python application with a graphical user interface that automates the full *molecular docking* workflow (ligand/receptor preparation, docking, interaction analysis) between pesticide compounds and *Apis mellifera* receptors, integrating MGLTools, AutoDock Vina, and GNINA.
- Implemented post-docking analysis modules (RMSD comparison, interaction detection, web-based result visualization) to support large-scale risk-assessment studies on bee mortality.

EDUCATION

University of Naples "Parthenope"

Naples, Italy

Master's Degree in Applied Computer Science | Machine Learning & Big Data

April 2026

Grade: **110/110 Summa Cum Laude** | Thesis: "*Feature-Driven Bias Detection: An Association Rule Mining Approach to Analyze Feature Importance*" | [Source code](#)

University of Naples "Parthenope"

Naples, Italy

Bachelor's Degree in Computer Science

December 2022

Grade: **110/110 Summa Cum Laude with Special Mention** | Thesis: "*Molecular interactions & automation of molecular docking*" | [Source code](#)

Liceo Scientifico "Enrico Medi"

Cicciano (NA), Italy

High School Diploma in Scientific Studies

Grade: 84/100

PUBLICATIONS & CONFERENCES

- L. D'Errico, R. Esposito, M. Matarese, V. Mele, **A. Mungari**, M. Roscica, M. Staffa. *Emotivation in Human-Robot Interaction for Affective Behavioral Adaptation*. In: *Social Robotics + AI (ICSR+AI 2025)*, LNCS vol. 16133, Springer, 2026. [\[DOI\]](#) [\[ResearchGate\]](#)
- Attendance*: The 2nd Italian Conference on Big Data and Data Science (ITA DATA), Naples, Sept. 2023.

HONORS & AWARDS

- Winner – BOOM Gen AI Innovation Sprint (CRIF), June 2026**: led a 6-person team as **Project Manager**, winning the **KPMG** challenge with *SpendWise*, an AI-agent-powered procurement intelligence platform.

SKILLS

- Programming Languages**: Python, C, C++, SQL, Java, JavaScript, Swift (iOS).
- Machine Learning & Data Science**: PyTorch, scikit-learn, CatBoost, mlxtend, Pandas, NumPy, Generative AI (Diffusion Models), Explainable AI (Counterfactuals, BoCSoR), Association Rule Mining (FP-Growth), Fairness Metrics, Jupyter Notebook.
- GPU & Parallel Computing**: CUDA, Multiprocessing.
- Tools & Methods**: Git, Conda, Python venv, $\LaTeX$, Project Management.
- Systems**: Linux, Windows, macOS.
- Languages**: Italian (Native), English (C2 | British Institutes ESOL International).

ADDITIONAL INFO

Driver's License: AM, B (own vehicle) • **Courses**: Apple iOS Foundation Program • **Other**: Youth Basketball Referee, FIP Campania (2016–2017).